MSE 360 Final Exam Review

Chapter 3 – Chemical Reaction Kinetics

Problem 1. A certain reaction has the following general form:

$$aA \rightarrow bB$$

At a particular temperature and C_{A0} = 0.100 M, the concentration vs. time data were collected for this reaction, and a plot of $1/C_A$ vs. time resulted in a straight line with a slope value of 4.15×10^{-3} L/mol·s.

a) Determine the rate law, the integrated rate law, and the value of the rate constant for this reaction.
b) Calculate the half-life for this reaction.
c) How much time is required for this reaction to be 75% complete?

Problem 2. The rate of a certain reaction doubles on increasing the temperature from 290 to 300 K. Evaluate the activation energy.

Chapter 4 – Transport Kinetics

Problem 1. Hydrogen gas splits into atomic hydrogen after adsorption onto a palladium (Pd) membrane and then diffuses via solid-state diffusion through the membrane. The atomic hydrogen diffusion can be measured by exposing one side of a Pd membrane to deuterium gas and measuring how long it takes for the deuterium to begin showing up on the other side of the membrane. Deuterium is a "heavy" isotope of hydrogen gas, and hence can be easily differentiated based on its mass.

Consider a Pd membrane of thickness, δ, as shown in **Figure 1**, which is initially devoid of any deuterium (D). At time $t = 0$, one side of the membrane is exposed to a D_2 concentration, C_D^0, while the D_2 concentration on the other side of the membrane is effectively maintained (by pumping) at zero for all time.

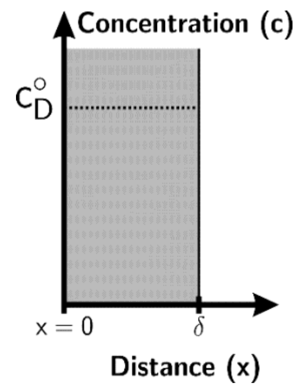

Figure 1

a) Several different kinetic processes can be rate limiting in this system: give at least three examples.

b) Assuming that solid state diffusion of D through the membrane is rate determining, draw a diagram illustrating the D concentration profile across the membrane at five times: $t = 0 < t_1 < t_2 < t_3 < t_\infty$.

c) Assuming that solid-state diffusion is rate-determining, provide the boundary conditions (B.C.'s) and the initial condition (I.C.) for this transient diffusion problem.

B.C. 1:

B.C. 2:

I.C.:

d) By employing your boundary conditions and initial conditions to evaluate this transient diffusion problem, you arrive at the following equation which describes the amount of time it takes before an appreciable amount of deuterium begins to appear at the downstream face of the membrane. This time is referred to as the "lag-time," t_L:

$$t_\mathrm{L} = \frac{\delta^2}{6D}$$

For a 51 μm thick membrane at 401 °C, you measure t_L to be 63 min. Based on this result, what is *D* (cm^2/s) for deuterium in Pd at this temperature?

e) Similarly to the reaction rate constant, (k), the diffusion coefficient (D) captures the thermally activated nature of diffusion. It is defined as $D = D_0 exp\left(\frac{-\Delta G_{act}}{RT}\right)$ where D_0 is a pre-exponential factor and ΔG_{act} is the activation energy for diffusion.

In the deuterium experiment, if you increase the temperature to 501 °C and t_L decreases to 21 min, estimate the activation energy (kJ/mol) for deuterium diffusion in Pd. (Hint: use your answer to part d).

Problem 2. Equation 4.40 in the text provides the solution for the transient diffusion of a "thin layer" of material between two semi-infinite bodies:

$$C_i(x,t) = \frac{N_i}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

This thin-film solution yields a Gaussian profile that gradually broadens as a function of time.

Derive a mathematical expression quantifying how this Gaussian profile "broadens" as a function of time. Use the peak width at half-maximum (i.e., the distance between the two points where the concentration is one-half of its peak value at any given time) as the definition of peak breadth.

Chapter 5 – Gas-Solid Kinetic Processes

Problem 1. During the calcination of limestone, calcium carbonate ($CaCO_3$) is decomposed into lime (CaO) and carbon dioxide:

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$$

a) For this solid-gas decomposition process, either the reaction at the surface or the gas-phase diffusion of CO_2 away from the surface could be rate limiting. Provide sketches illustrating how the CO_2 pressure would vary between the surface of the $CaCO_3$ and the bulk atmosphere of the furnace for each of these two cases.

a) Assume that the rate of calcination is controlled by the rate of diffusion of CO_2. Assume that the CO_2 leaving the surface diffuses through a stagnant gas diffusion layer of air 1 mm thick and that the CO_2 pressure in the bulk of the calcining furnace is effectively zero. Calculate the rate of decomposition of limestone in units of moles/cm^2·s (e.g., the flux) at 1000 K given that $D_{CO_2} = 4.7 \times 10^{-4}$ m^2/s at 1000 K and that the pressure of CO_2 at the surface is 0.054 atm.

Chapter 6 – Liquid-Solid and Solid-Solid Phase Transformations

Problem 1. Provide a sketch of system free energy (ΔG) versus nucleus size (r) during a solid-state phase transformation. Include four curves on your sketch:

a) A curve indicating how the **interfacial energy** of the system varies with nucleus size
b) A curve indicating how the **volumetric energy** of the system varies with nucleus size
c) A curve indicating how the **total free energy** of the system varies with nucleus size
d) A curve indicating how the **total free energy** of the system varies with nucleus size **for the case of heterogeneous nucleation where $\boldsymbol{f(\theta) = 0.5}$.**

For curves c) and d), **indicate $\boldsymbol{r^*}$ and $\boldsymbol{\Delta G^*}$.**

Chapter 7 – Microstructural Evolution

Problem 1. Solid silver has a density of 9.3 g/cm^3, an atomic mass of 107.87 g/mol, a liquid-solid surface energy of 1.3 x 10^{-5} J/cm^2, and an enthalpy of melting of 11.3 kJ/mol. Given that silver has a bulk melting point of 962°C, calculate the estimated melting point of a silver nanoparticle with a radius of 7.0 nm.